REPORT E-Commerce Sales Analysis

REPORT E-Commerce Sales Analysis Dashboard

Overview

This project focuses on analyzing ecommerce sales data using Excel to build an interactive dashboard. The dashboard integrates multiple KPIs, charts, and customer/product-level analysis to provide actionable insights for decision-making. It helps in understanding sales performance, profitability, and customer behavior across different dimensions such as time, region, and product category.

Introduction

Ecommerce businesses generate vast amounts of data daily. To remain competitive, companies must transform this data into meaningful insights. A sales analysis dashboard enables business managers to monitor performance, identify top products and customers, and evaluate profitability.

This project leverages Excel tools such as Pivot Tables, Pivot Charts, Slicers, and Conditional Formatting to create a dynamic dashboard for ecommerce data analysis.

Objectives

- To analyze ecommerce sales data and identify business trends.
- To design an interactive dashboard for better decision-making.
- To evaluate sales, profit, and product performance across categories and regions.
- To track top-performing customers and products.
- To establish KPIs for measuring business growth and efficiency.

Dataset Description

Dataset

The dataset contains **22 columns** and multiple records of ecommerce transactions. Key fields include:

- **Order Details**: Order ID, Order Date, Ship Date, Ship Mode
- **Customer Details**: Customer ID, Customer Name, Segment, Country, City, State, Region
- **Product Details**: Product ID, Category, Sub-Category, Product Name
- **Sales Metrics**: Sales, Quantity, Discount, Profit

This structured dataset forms the foundation for analysis, aggregations, and dashboard visualizations.

Dashboard Features and Visuals

The dashboard integrates multiple visual elements:

- **Combo Chart**: Monthly Sales vs Profit trends
- **Waterfall Chart**: Profit contribution by product categories
- **Pie Chart**: Sales distribution across categories
- **Map Chart**: State-wise sales representation
- **Bar/Column Charts**: Top 5 customers and top 5 products
- **KPI Cards**: Showing Sales, Profit, Quantity, Orders, and Profit Margin

Interactive features include slicers and filters for dynamic exploration of data.

KPIs Used

Sales –Measurestotal revenue generated.

Profit –Tracksbusiness profitability.

Quantity Sold–Indicates demand for products.

Number of Orders– Shows customer engagement.

Profit Margin –Evaluates efficiency of pricing and cost control.

These KPIs provide a comprehensive view of business performance.

Insights from Dashboard

- Technology contributes the **highest profit**,whileFurniturelagsbehind.
- Sales distribution is fairly balancedacross **Furniture (32%), Office Supplies (31%), and Technology (36%)**.
- **Top 5 customers** account for a significant share of sales revenue.
- Certain states (e.g., California, New York) outperform others in sales.
- Sales and profit show **seasonal peaks** in certain months (e.g., March, November).

Referral Image

Conclusion

The Ecommerce Sales Analysis Dashboard successfully highlights critical aspects of business performance. It provides managers with a consolidated view of sales and profit trends, customer contributions, and category performance. This aids in strategic planning and helps identify growth opportunities.

Future Enhancements

- **Automating data refresh with live ecommerce data.**
- **Integrating with advanced BI tools like Power BI or Tableau.**
- **Adding predictive analytics to forecast future sales and profit.**
- **Including customer segmentation and lifetime value analysis.**
- **Expanding dashboard for multi-country and multi-channel sales.**

Appendix: Charts Used

1. **Combo Chart – Monthly Sales vs Profit**
2. **Waterfall Chart – Profit contribution by category**
3. **Pie Chart – Category-wise Sales Share**
4. **Map Chart – State-wise Sales Performance**
5. **Top 5 Customers – Bar Chart**
6. **Top 5 Products – Column Chart**
7. **KPI Cards – Sales, Profit, Quantity, Orders, Profit Margin**

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